

Media Engineering and Technology Faculty
German University in Cairo



Psychiatry Counseling in the Metaverse I (VR)

Bachelor Thesis

Author: Mark Emeel Sroor
Supervisor: Dr.Wael Abouelsaadat
Submission Date: 11 June, 2022

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This is to certify that:

- (i) the thesis comprises only my original work toward the Bachelor Degree
- (ii) due acknowledgement has been made in the text to all other material used

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11 June, 2022

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Thank you to my supervisor Dr. Wael Abouelsaadat. Your guidance, wealth of knowledge and genuine interest were paramount in making this thesis and project a reality.

To my family, I thank you for your love, trust and support that you have provided and continue to provide. I love you with all my heart and soul.

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To my best friend Michel Gamal, thank you for being a sturdy pillar of support that I could count on.

Abstract

Mental illnesses are quite common in our communities. However, the means to deal with these illnesses are lackluster. People with mental health issues rarely seek adequate treatment, and those who seek it struggle to find it. The pandemic Covid-19 made matters worse in making mental health problems worse, and limiting the ability to seek help even further. Both traditional therapy and online therapy could stand to be improved. In this thesis, we proposed and implemented a system for psychiatry in the metaverse using VR. The system allows for a psychiatrist to meet with their patient in a virtual office. The system amplifies the experience by adding extra tools and experiences that would be cumbersome or expensive to achieve by other means. This thesis focuses on the VR subsystem, and how it implemented a seamless form of interaction, not only between both the psychiatrist and the patient, but also between each user and the world around them.

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Chapter 1

Introduction

1.1 Motivation

The world is learning how important it is to care for your mental health. The World Health Organization (WHO) reported in October of 2021 an increase in global mental health promotion and prevention programmes, going from 41% of member states to 52% in 2020. [1]. Nevertheless, a mere fraction of those affected by mental illnesses receive the adequate treatments and/or medications. The Egyptian Ministry of Health conducted a survey in 2018 over a random sample of 22,000 families. The survey reported that 25% of Egyptians suffer from one form of mental health problem. Then, it is even more concerning that only 0.4% of the 25% are receiving the adequate treatment and/or medication. [2]

Some people refuse to get medicated because of negative stigma surrounding mental illness. Some people downplay the severity of the issues. Some people can not physically find a therapist that is near them, or they are disabled in a way that prevents them from locomotion. Others who are living abroad find it difficult to relate to the therapists of the foreign country and would prefer to talk to someone from their homeland.

The Covid-19 pandemic exacerbated problems even further. The WHO reported in March of 2022 that the pandemic lead to a 25% increase in anxiety and depression worldwide. This increase was due to many factors. The social isolation not only caused loneliness, but also constrained people's ability to work, and to receive support from loved ones when needed. Fear of infection, loss of jobs, and fear of suffering for oneself and for loved ones lead to great grief in the world. [3]

All the while, it was becoming more and more difficult to receive the necessary care. The WHO pointed out in the same article that this increase in the prevalence of mental health problem coincided with the disruptions that affected the mental care systems due to the pandemic, leaving a large gap in the care of those who need it most. By the end of 2021 and going into 2022, the situation had somewhat improved, but much of

the population still struggles to get the support they need for pre-existing and newly developed medical conditions from the pandemic.

The pandemic highlighted our gap in mental care systems, and signalled the need for better, more accessible tools to help those in need when they need it.

The metaverse stands to solve many of the problems of traditional therapy and online therapy. In terms of traditional therapy, it allows for a cost effective way to meet psychiatrists, and even change psychiatrists if the ones found were not suitable or adequate. It negates locomotion time, meaning that people can get help right when they need it, and not be limited to psychiatrists that are in their immediate vicinity. It also allows for better privacy, as the help could be received right from the patient's bedroom.

In terms of usual online therapy over calls, the metaverse can also deliver much needed improvements. Online calls reduce people to pixels on a screen, often resulting in patients, as well as their psychiatrists, to feel disconnected. The calls feel disingenuous because they lack "real connection". Online therapy can still be effective, and this thesis covers studies that prove its worth, but we can still improve upon it.

1.2 Problem Statement

Online therapy is a great alternative to face-to-face therapy. However, it has many shortcomings. The aim of this project is to develop a therapy application for a psychiatrist to meet with his patients. The system intends to shore up the shortcomings of online therapy over conferencing app as well as traditional therapy. Moreover, it intends to augment the psychiatry session with experiences that could not be achieved without the use of VR.

1.3 Thesis Overview

This thesis is divided into 4 chapters. In chapter 1 we discussed our motivation for this project, and the problem statement we aim to solve. In chapter 2, we provide some relevant background info to be known for this thesis, and discuss prevalent previous works that delved into similar problems we faced. Chapter 3 includes our descriptions of our implemented system. It has our design goals, system overview and design, and the implementation details. Finally, chapter 4 concludes this thesis with what was achieved, the limitations of the system, and proposals for future work for further enhancements on the system.

Chapter 2

Literature Review

2.1 Background

2.1.1 Virtual Reality

Virtual Reality (VR) is a computer-generated environment in which the user is immersed in a digital world, and is usually allowed to interact with the objects around them in said world. Nowadays, this is usually achieved with Head-Mounted Displays (HMD)s. An HMD displays a different image for each eye to form an overall stereo scene. It is typically tracked, either with dedicated IR-sensors (like the HTC-Vive headset), or using inside-out trackers that work with visible light (like the Meta Quest 2 seen in figure 2.1). VR is usually associated with entertainment. However, it is projected to be used in more domains, such as in healthcare, education, and workforce development.[\[4\]](#)



Figure 2.1: The Meta Quest 2, an example of an HMD, and it is the HMD we were using.

2.1.2 Metaverse

The metaverse is a term used to describe a world where people interact and live their lives over the internet using HMDs. This world would make use of virtual, augmented and mixed reality technologies. The term originates from the 1992 science fiction novel *Snow Crash*. However, in this paper we'll be using the term to refer to the metaverse proposed by Mark Zuckerberg, co-founder of Facebook and its parent company Meta Platforms.

2.1.3 Traditional Therapy

Traditional therapy is used in this paper to refer to face-to-face therapy.

2.1.4 Online Therapy

Online Therapy refers to any form of therapy that is performed over the internet. It includes therapy that is provided over telephone, video-conferencing apps, and/or text messaging apps. This can also include guidance from a counsellor, or using automated experiences that do not involve a counsellor.

2.1.5 Exposure Therapy

People tend to avoid situations they fear. This avoidance can reduce feelings of fear in the short term, but actually increases them over the long term. In order to break the cycle, counselors guide their patients through those feared situations in a safe environment, proving to them that they can be safe. This treatment method is called Exposure therapy, and it is used to treat different illnesses, such as phobias, and anxiety disorders.

2.1.6 Working Alliance

A term that refers to the relationship that builds between therapists and their patients.

2.1.7 Human-Computer Interaction

Human-Computer Interaction (HCI) is a multi-disciplinary field of study which focuses on the design of optimal interaction between humans and computers in terms of efficiency and ease-of use.

2.2 Previous Works

2.2.1 Online Therapy

This section summarizes the findings by Barker & Barker [5], Jardine et al [6], and Békés et al [7] about online therapy. These papers were especially considered because they cover online therapy during and after the events of the Covid-19 pandemic that happened in 2020, and whose effects are still felt today in 2022. Internet-based therapeutic services have been available for decades now. And even though research has showed time and time again that it can be effective, online therapy is not popular amongst counsellors. Most of them prefer Face-to-Face therapy. They feel that online therapy lacks important, non-verbal physical cues. These cues can convey a wide array of information, and they do not translate to a digital medium as well as they require. These cues include posture, gaze, hand and leg actions and positions, and many other signs we tend to convey without us even noticing. For that reason, therapists prefer to have their patients in their offices, where they can get a much closer, keener look at them.

The therapists' apprehensiveness has hindered the scientific research into the efficacy of online therapy. These preconceived notions caused biases in the evaluation of the methods, and online therapy was never regarded as a viable method of therapy. Patients, however, do not share the same sentiment. Patients generally have high expectations of online therapy, and this optimism makes the patients more ready to change.

In March 2020, the Covid-19 pandemic had forced all of us to quarantine in our homes. People were forced to work from home. Firms and institutes had to find different ways to provide their services remotely. There was a decline in mental health during the quarantine. There were families that were separated, in some cases across countries apart, while other families quarantined together were confronted with the realities of their fractured relationships. Kids were forced to stay inside and learn over Zoom (a video conferencing app that was popular at the time), causing general distress for the parents who, all of a sudden, did not have any time for themselves in the day, and had to be more involved in their kids learning process.

Outside the home, jobs were lost due to the switch to remote working conditions, while others were heavily overworked because their colleagues were laid off. Those who had to work online realized that they were not at all prepared for it, as they lacked the proficient training to carry out their jobs in a remote setting. 'Zoom fatigue' was a term used to describe the general discomfort people faced after hours and hours of talking and working over online video-conferencing apps. As it is a fairly new problem, there is not much research into the reasons for the fatigue, but it research done by Fauville et al suggests [8] that some reasons could include that most video conferencing apps have a window where you view yourself, which can be uncomfortable over long hours of use. It is also uncomfortable as you are expected to keep eye contact for an extended period of time, and that it is just tiring for some people to look at a screen for that long.

The pandemic also had a toll on healthcare workers. They were overworked and understaffed. Barker and Barker [5] cite that the most common calls to the National

Domestic Violence Hot-line in the USA between March and April in 2020 were from healthcare workers, who reported abuse by people who claimed that they were purposely infecting people with the virus. The workers also reported the fear of carrying over the infection back to their families, and that they could not handle the number of sheer people dying in hospitals everyday. The decline in mental health in turn caused a stress over counselors, who now were forced to continue their practice online as their only method. And even though they faced a lot of technical difficulties during their work online, it was the first time online therapy was seriously used, not out of preference, but out of necessity. Researchers were finally given the chance to gauge the efficiency of online therapy.

Indeed, research found that online therapy is, at least, just as effective as Face-to-Face therapy. It is certainly better than no therapy whatsoever [5][6][7]. The counsellors doubts resolved over time as they gained more experience with the medium, and they had a chance to finally see the benefits online therapy provides.

First, the most glaring benefit is that it completely cuts commuting time. This is not trivial, and can be the main cause that patients would not seek help. Some patients simply do not have any counsellors withing their vicinity. Even if they do, they can be limited in their choices, and it is then left to chance whether they click with the available counsellor or not. Some patients can not leave their home for health reasons, including actual physical conditions, or because of their mental illnesses such as social anxiety or depression. This immediacy shortened the time between inquiry and intake, bringing help to patients the moment they need it, and increases the number of patients that each counsellor can help on a daily basis. Additionally, the fact that the patients receive the treatments in their homes, they have an easier time immediately transferring skills they learn during the session from practice to application.

It also provides some anonymity, which is needed for patients with low self-esteem, like patients with high levels of body shame, self-consciousness, and social anxiety. On-line therapy gives a degree of space and safety which are needed for the initial steps of treatment. It also provides a sense of control over the session, as they are free to end the call at anytime; this control is important for patients who yearn for it in their day to day lives, such as victims of abuse.

The researches, however, reported that therapists found they were more distracted in online sessions, and there was a general lack of privacy at home for themselves and patients. They also reported their worries regarding the technologies used and the technical issues they faced, but these issues were attributed to the stress the services were facing due to the unprecedented demand of them.

Another interesting detail noted by Jardine et al [6] is that factors extraneous to the intervention itself affected the treatments; factors including personal motivation of the patient, lack of computer/technical skills, fatigue from using computer at work, a desire for human contact. Also, it was found that the patient's belief in the effectiveness of the treatment methodology greatly affected the outcome of the treatment. Therefore, the design and evaluation of online interventions has been an area of interest in HCI research [6], because of the complex design requirements that need to match the sensitive

nature of the interventions and the content involved. The easier and more streamlined the experience, the better the patients would respond to the treatments.

Most importantly, the papers did convey that online therapy might be better suited for certain mental illnesses over others. Online therapy may be unsuitable for clients with paranoid, rigid and avoidant traits; clients who suffered from childhood abuse; and clients with symptoms of emotion dysregulation and dissociation [5].

These findings indicate that online therapy may be more than just an alternative to traditional therapy. As counsellors gain more experience with it, hopefully they could make better use of the benefits it provides.

2.2.2 Working Alliance

This section summarizes the findings of Cook, M.S & Doyle [9], and Berger [10], regarding working alliance.

Nowadays, a lot of randomized controlled trials (RCTs) have proven the efficacy of Internet interventions for a wide range of mental disorders. [10] Studies directly comparing Internet interventions with face-to-face therapy report similar outcomes across various mental disorders (e.g., anxiety disorders, depression) and health concerns associated with bodily symptoms (e.g., tinnitus, sexual dysfunction). There are also a few long-term follow-up studies showing lasting effects over as much as 5 years' post-treatment. Most of the studies showed that the best outcome comes from a blend between a Web-based self-help program with minimal but regular therapist contact through secured e-mail. Studies also show a superiority of guided interventions versus unguided, automated programs in terms of efficacy, compliance, and attrition rates.

Berger [10] discusses guided self-help Treatments. Most of the studies demonstrating the efficacy of Internet interventions investigated guided Internet based cognitive-behavioral treatments (ICBT). In guided ICBT, the presentation of a web-based self-help program is combined with minimal but regular, usually weekly therapist guidance. Guidance provided by a therapist can take different forms and may differ in terms of the amount of time devoted to the each client. Yet, many studies involve rather low-intensity guidance with nearly 10 minutes of time spent by a clinician per client and week. Guidance often consists of weekly scheduled written feedback by a therapist and the possibility for clients to ask questions. The main purpose of the guidance is usually to explore and reinforce the participants' independent work with the self-help guide.

Working alliance can be measured from the perspective of clients, therapists, and independent observers, but research has consistently shown the strongest correlation between clients' view of working alliance and outcome. While it appears that type and severity of clients' pretherapy symptomatology is unrelated to establishment of alliance [9]. Therapists in guided ICBT often apply some therapist behaviors such as; conveying understanding, empathy, warmth and concern for the participant's suffering and situation [10]. The cumulative results of research on Internet relationship development indicate

that relationships are frequently formed online and that they are not less strong than their face to face counterparts. Moreover, many, if not most, of the relationships developed on the Internet occur through text-based mediums which is usually considered the cheapest mode of communication online. It is likely that many of these same elements (e.g., empathy, support, and a readiness to self-disclose) that develop informally will encourage deep and intimate relationship formation in online environments, enabling therapists and clients to embrace working alliance in therapy[9].

A review conducted by Berger [10] showed that regardless of communication modalities, diagnostic groups and amount of contact between clients and therapists, client-rated alliance scores were high, roughly equivalent to alliance ratings found in studies on face-to-face therapy. This suggests that at least from the perspective of the clients, a positive alliance can be established in Internet interventions independently of delivery modes and communication modalities. Still another consideration is that specific “alliance measures” should be developed in order to assess specific characteristics of the online relationship.

Regarding the concerns to Internet interventions; issues about confidentiality, privacy, and reliability of technology usually rise on the top when discussing the pros and cons of this type of therapy.

2.2.3 Presence and Embodiment in a Virtual World

Steed et al [11] wrote their paper at the precipice of consumer virtual reality becoming widely available. Steed et al wanted to assess best practices in not only providing the best sense of presence and embodiment in VR, but also to not induce motion sickness in users. They designed an experiment to assess what exactly goes into making a user more present in VR. The experiment was using simple Samsung Gear VR and Google Cardboard, which were the widely available ways to engage with VR content at the time. The experiment would put the users in a scenario where they sat down and watched a singer perform in a bar. They would then answer questions regarding their experience. The scenario had multiple variations: whether the user had or did not have a self-avatar; whether the singer looked at and engaged in eye contact with the user or not; whether the user was asked to tap along to the music or not. The study had three hypotheses in regards to presence. The first hypothesis was that having a self-avatar would have a positive impact on self-reported sense of presence. Second, having eye-contact with the singer would lead to a better sense of immersion. Finally, engaging with the content in VR by tapping along would lead to a higher sense of immersion.

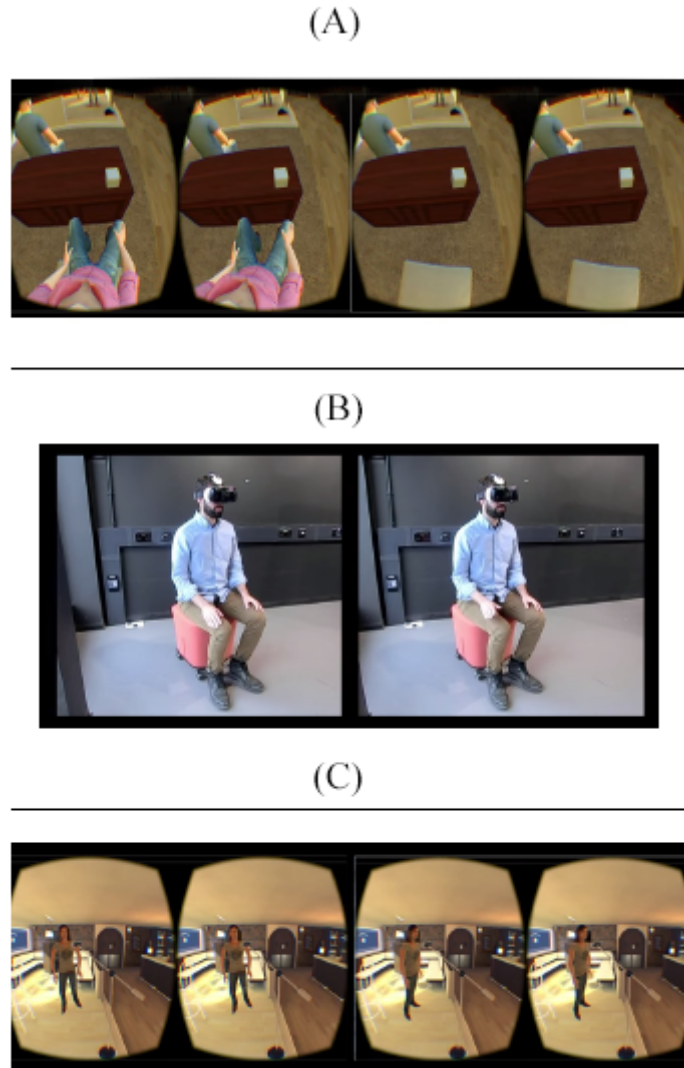


Figure 2.2: The scenarios users experienced in Steed et al [11] experiment to test what affects their sense of presence in VR: (A) Self avatar vs. no self avatar, (B) User asked to tap along at the start vs. not, and (C) Singer looking at user vs. singer ignoring user.

The study reported a positive impact of having a self-avatar. Having an avatar to represent that you are actually part of the world was found to enhance the sense of presence in the virtual world. The second hypothesis of having the singer engage in eye contact with them was reported to have no effect on presence. Steed et al hypothesized that because the singer was performing, the user was not expecting to engage with the singer, and thus the eye contacts were thought of as random looking around by the model of the singer. Also, due to the lower visual fidelity of the used devices, and the distance at which the singer was at from the user, it was hypothesized that the users could not tell that the singer was having any eye-contact with them. The third hypothesis of having

the user tap to the music was actually reported to have a negative effect on immersion, which as Steed et al point out was to be expected in hindsight. Samsung Gear VR and Google cardboard do not provide any motion tracking for the users body. Having the user engage in any physical movements and having their avatar not follow such movements is a reminder that the body they see is not their own.

This paper highlights some interesting notions. First, having an avatar is really important for the sense of presence in virtual reality. Second, it is really important for the movements of the avatar to match the movements of the user. Finally, even though the study reported eye contact to not be an important factor, it can be claimed that the singer was a virtual character and not another person that they could interact with.

Sharing a Virtual Space and its effects on presence

VR has a problem that most of its experiences do not involve other people. Fribourg et al [12] and Hosoya et al [13] tested out the sense of embodiment and immersiveness when other users share the same space. They do not have to be in the same physical room, but they still feel like they are together. This has the potential to connect people who can not usually meet for different reasons, such as distance, health, or even just convenience.

Fribourg et al explored the influence of sharing an environment with another user in VR on the Sense of Embodiment (SoE) and presence. SoE is how much a user claims ownership of their avatar. Presence refers to how much a person feels like they are present in a virtual world. The study mentions other works and researches into the SoE and presence in VR. They conducted an experiment where users were placed in a VR scene playing a version of a whac-a-mole game. The users had humanoid avatars that represented themselves and other users around them to enhance the SoE. The main objective was to determine how having other people around in VR enhances presence, and maybe their SoE.

The experiment consisted of 3 scenarios: playing the game alone, playing the game in front of a mirror, and playing with another user in VR.



Figure 2.3: The different scenarios posed in Fribourg et al's experiments. From left to right: experimental conditions Alone, Mirror and Shared; Physical setup of the experiment.

On user performance and engagement, the study noted that the players were more "efficient" in performing their tasks when sharing the environment with another player, meaning they were more immersed and engaged in their task. This was deduced by performance metrics set by the study, but also the user feedback after.

On body ownership, the study had an interesting scenario where dangerous spikes were placed around the game. It was noted that users placed their hands higher when spikes were present. However, the study could not discern whether this was done out of fear for their avatars (and by extension themselves) or if it was just to avoid collision. Nevertheless, user feedback included that they felt strange when they moved their arms through the spikes. It took them several seconds to be at ease with them and realize they could not be harmed. The study also noted that the presence of a mirror did not increase body ownership. The study hypothesizes that the mirror could have been distracting, or makes the users notice animation glitches in the avatar, even though such a concern was not reported by the users.



Figure 2.4: The dangerous spikes that were tested to see if users were fearful for their virtual self.

The study was limited by the small sample size of only 20 participants that were all male. Also, the results of the study where increased presence were not clearly attributed to presence of another user in VR, but could have been due to the competitive nature of the environment of the experiment. Moreover, participants reported that being forced to be seated, and not being able to explore the room enforced that it was a virtual area and cut the immersion.

Hosoya et al [13] proposes their design of HyCLASS, a VR environment that allows student to come together and interact. It can be used to perform virtual experiments and carry out different tasks. The system adds the ability to create new objects and modify their properties. The idea for a VR classroom is motivated by helping students more intuitively understand school material by having students interact with the material in 3D space and observing objects in multiple viewpoints.

As noted by [13], for optimal immersion and sharing virtual space, there need to be a set of conditions met:

1. The users should share the same environment.
2. When an object is manipulated, by the system or any user, all users should see the same change.
3. Avatars need to be realistic, humanoid models.
4. Real-time conversation should be present.

2.2.4 VR Therapy

VR can be very immersive as noted above, and can evoke a sense of embodiment in the virtual world. This embodiment not only helps the patient feel close to the counsellor as if they are actually in the same room. Moreover, this immersiveness can be used as a new learning tool. Patients can go through simulated scenarios as a means of practice for the real thing. For example, a person who is afraid of public speech can give out his speech in VR to a huge audience beforehand. As pointed by Freeman et al[14], Patients know that a computer environment is not real but their minds and bodies behave as if it were real. This phenomenon makes them feel enough to go through these experiences, and real enough for those experiences to matter. This is really interesting because doctors can exploit such an experience to their advantage. By showing a patient that what they perceive does not have to be necessarily real, counselors can teach them that what their brain tells them can also be false. Patients can start then to try and change their perception with the knowledge that they *can* be wrong.

These experiences can be automated, and then they can be repeatable for whenever the patient needs them. VR allows tight control over the simulations which makes it so therapeutic strategies can be precisely implemented, and that their quality would remain consistent over multiple runs. It also includes experiences that are inconvenient for the counsellor to repeat for each patient, like fear of heights and going to tall buildings, and experiences that are too expensive to be repeated, like for patients who are afraid of travelling by plane. These simulations can be graded in difficulty and experienced until recovery is achieved.

VR has the potential to not only be used in treatment, but also in diagnosis of mental illnesses. Freeman et al [14] point out that humans tend to be subjective in the views. More importantly, their memories can be very fallible, and can warp a situation to how they would like to remember it, as opposed to what actually happened. With VR technologies, the counselor can be directly present with the patient in their feared scenario, giving them the ability to directly diagnose their patient in real-time.

So far, the most common use of VR in therapy has been to simulate exposure therapy [14]. This method of treatment has already been used in treatment of many ailments,

such as persecutory delusions [15] [16], Post-traumatic stress disorder (PTSD) [17]. It has been used to treat body dysmorphia [18], and it has even been used to treat cravings, such as food cravings [19]. Finally, we look into an example of using VR to treat fear of public speaking [20].

Treatment of Persecutory Delusions

Both Freeman et al [15] and Pot-Kolder et al [16] looked into persecutory delusions and paranoid ideations. These are disorders where patients believe that other people are out there intending to harm them. Approximately 70% of patients with schizophrenia have this type of psychotic experience, which in turn leads to social withdrawal, suicidal ideation, and hospital admission. These unfounded beliefs are maintained by the defence, safety-seeking, behaviors of the patients. Patients would avoid other people, avoid eye contact, remain alert, and attribute the absence of harm to their constant vigilance, which reinforces that the reason they are safe is that they were vigilant enough in the first place. Traditional exposure therapy would be the mechanism used to treat persecutory delusions, but it has many limitations. First, the patients sometimes outright refuse to participate, because of the intolerable anxiety that comes with that experience. Second, the environment and reactions of others cannot be tightly controlled, leaving an element of variance that can propose dangers to the patient. Third, these sessions are done in between therapy, and thus the feedback on them is retrospective, which lowers the accuracy of the reports. All of these limitations can be surpassed in VR. VR therapy is tightly controlled; patients know they are safe, and the sessions are done with the presence of the counsellor so the reports are accurate. All the while, VR is more efficient at providing exposure therapy because all the different experiences can be had in one room, and the counsellor can easily switch between them to be tailored to the patient's needs.

Freeman et al [15] had already showed that they have used VR to assess paranoia and understand its causes, and that it is safe to use with patients with persecutory delusions. They conducted a pilot study first for those patients and showed large improvements in their conditions. They started using these findings to develop gameChangeVR, with a study in 2020 [21], and another in 2021 about their progress [22].

Both Freeman et al and Pot-Kolder et al conducted similar experiments in which they randomized a controlled trial to establish the effectiveness of VR Therapy in comparison with the traditional methods. They hypothesized that VR Therapy would lead to more improvement to the patients' conditions than with traditional methods. They also hypothesized that its effects would be maintained over time. The procedures included placing the patient in a social environment that is controlled by the counsellor. Patients use avatars that would reflect themselves, and interact with other avatars that can say pre-recorded lines. Their reactions over multiple sessions were recorded. Freeman et al added an element to their program about mental relaxation. Users were able to choose between different calm environments, like a beach, countryside, forest or lake, to practice relaxation techniques. These techniques were hypothesized to counteract fear of other

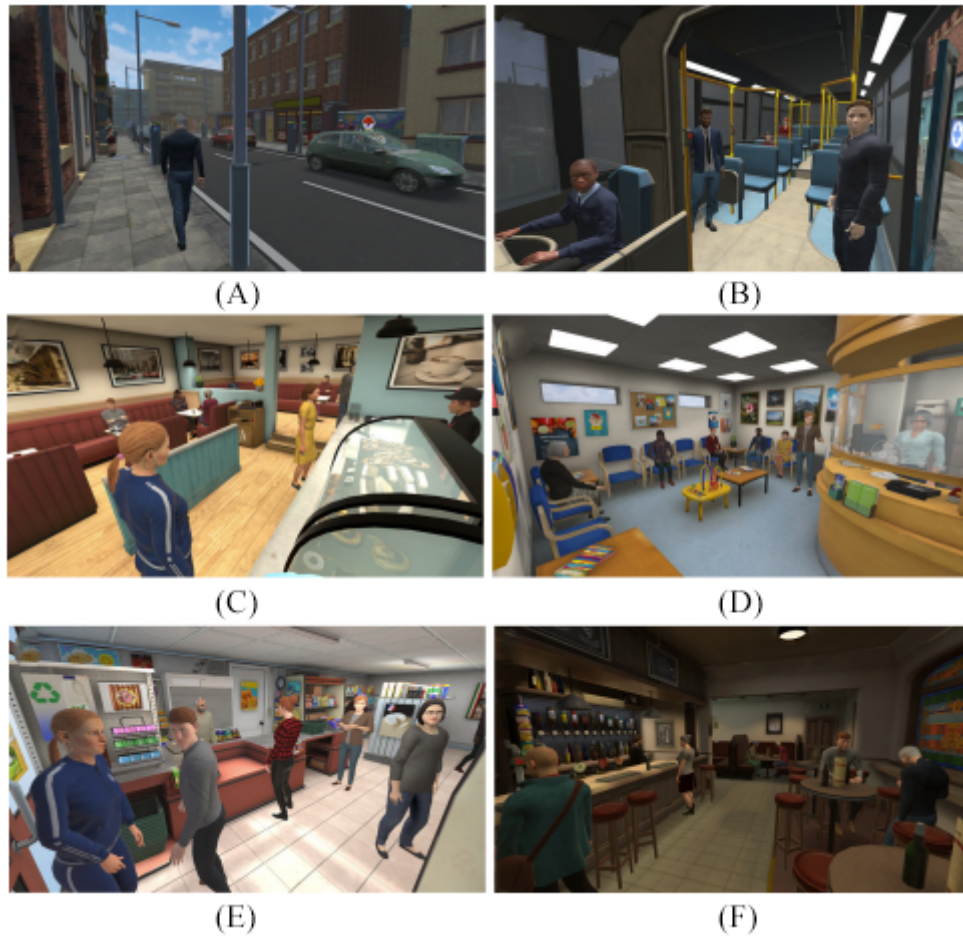


Figure 2.5: The six gameChange scenarios: (A) Street, (B) Bus, (C) Café, (D) Doctor's waiting room, (E) Corner shop, and (F) Pub to test assessment of persecutory delusions in different public scenarios.

people by having a calm mind. While Freeman et al has not fully finished their research at the time of this writing, Pot-Kolder et al reported large reductions in momentary paranoia and safety-behaviors during social interactions, which held up at the 6 month follow-up assessment. However, they conceded that VR was not significantly better than traditional therapy in outcomes, and that its effects are not as rapid. Their conclusion could be due to the small-time frame this study had taken place in. It was also, to their knowledge, the first randomized controlled trial of VR-CBT. Nevertheless, they still recommended VR exposure therapy as it is still more practical due to its extreme convenience.

Treatment of Post-Traumatic Stress Disorder

Reger et al [17] applied the principles mentioned above to treat PTSD in US military service members who were deployed in Iraq and Afghanistan. There is a history of mili-

tary members facing barriers to mental health care, including the stigma associated with seeking mental health treatment causing them to be seen as weak by leaders. VR has the chance to 'gamify' treatment, making it more accessible to them. The study divided participants into three groups, prolonged (traditional) exposure, VR exposure, and a waitlist group of those who had not received any treatment as a control group. Prolonged exposure is a manualized treatment for PTSD which involves psycho-education on PTSD and methods of dealing with it, and prolonged imaginal exposure to the traumatic memories. For the equipment, the study made use of an eMagin z800 HMD with an "inertia" cube orientation tracker for replicating the user's head movements. Patients navigated the scene with a Logitech joystick or a mini joystick attached to a mock M4 rifle. The users wore headphones for sound. The room was equipped with bass shaker speakers to simulate the vibrations of a battlefield. Finally, an EnviroScent Scent Palette was used to deliver olfactory stimuli. VR exposure placed the patient in a relevant VR environment, and patients would relive their memories that way. As the patient articulates their memory, the scene would be customized to match the memory.



Figure 2.6: The eMagin z800 headset [23].

After the experiments, the study concluded that while there were significant improvements in PTSD and depression after treatment with either traditional or VR exposure, neither of them were superior, and that there was no significant difference in the reduction of stigma to therapy. However, soldiers who received traditional therapy seemed to maintain the benefits acquired from therapy for a longer period of time compared to those who underwent VR therapy, but the study could not find why. They hypothesized that some patients were distracted or not immersed enough, which now we know makes sense. The quality of the HMD used in this experiment is subpar. It only provides a resolution of 800x600, which is quite low compared to HMDs we have today. For example, the Meta Quest 2 headset provides a resolution of 1920x3664. Also, the users did not have motion tracking, and only moved their characters with joysticks. Even when attached to a mock rifle, the joystick would never be sufficient enough to simulate actual presence.

Treatment of Weight Disorders

M. Ferrer-Garcia et al. [19] approached VR therapy differently. Instead of using VR therapy for exposure of fearful experiences, they used the fact that VR tricks the body and the mind that their presented scenario is the real thing. The study used VR to treat eating disorders by simulating real-life situations, and encountering food-related cues. The study posed an experiment of putting patients in 4 different scenarios: being presented with low-calorie food alone (in a kitchen) and with other people (in a restaurant), and high-calorie food alone and with other people.

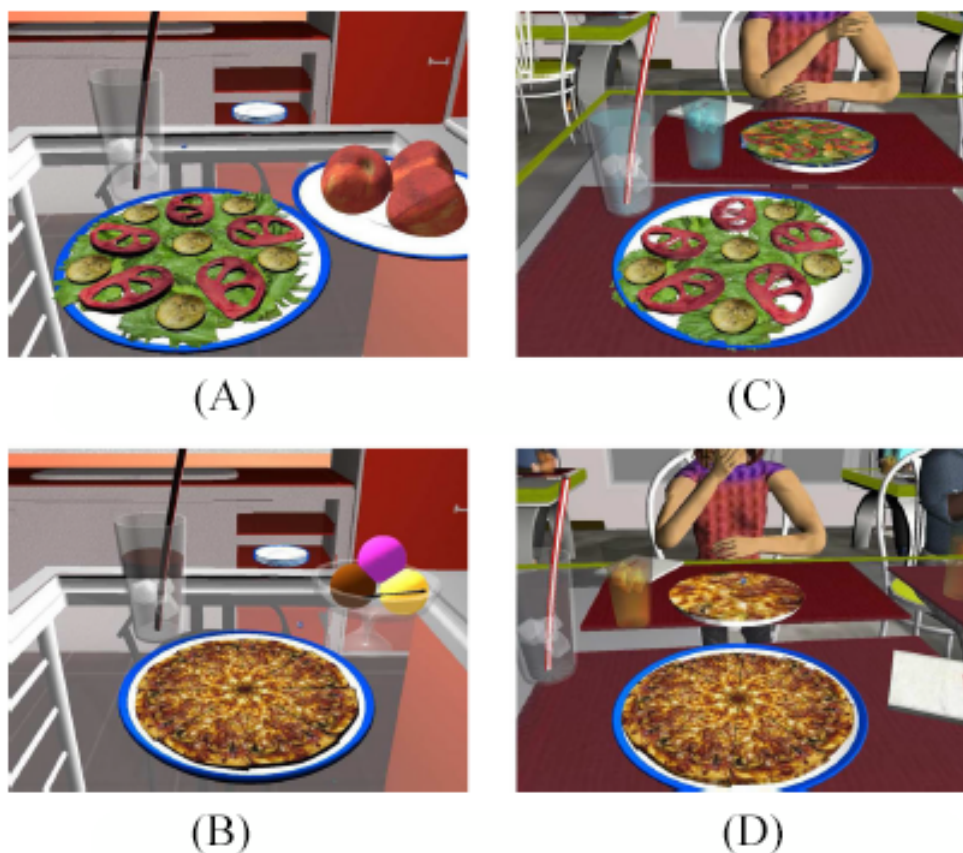


Figure 2.7: The 4 scenarios posed by M.Ferrer-Garcia et al: (A) Low-calorie kitchen, (B) High-calorie kitchen, (C) Low-calorie restaurant, and (D) High-calorie restaurant.

The study does not use an HMD, but rather a screen that produces a "3D" image and users would wear polarized glasses. For sound, they used noise-cancelling headphones. The experience was passive, meaning that users were not able to interact with the material. Even though this is not a VR setup we identify with today (as noted by Freeman et al[14]), the study reports patients experienced cravings for high-caloric food. Presence of other people was determined not an important factor. The fact that VR can induce cravings helps counsellors guide their patients with effective methods of dealing with those

cravings. However, the study mentions that it was limited by the fact that participants were all female college students. The study also did not specify instructions about food intake before the experiment, which could have skewed some results. They had all patients view the same food, while, generally, patients are exposed to their favorite snack or binge food.

In another study, Wiederhold, Malonado & Riva [18] explore the use of VR to treat weight-related disorders. They used VR to treat body dysmorphia, which is when a person has an incorrect view of one's body. Riva designed a VR app where patients with body dysmorphia can pick to use from 9 avatars that range from underweight to overweight. The discrepancy between the chosen avatar and the actual body of the patient was used to show the patient how wrong they are about their own views of their own body. The study references another paper by Perpiñá et al. where another approach was used. Patients can alter their human figure by enlarging body parts or making them smaller. Again, the same principle was applied in using the discrepancy mentioned above to treat the patients.

The main problem that faces the counsellors who treat patients with body dysmorphia is the lack of awareness that the patients show about the real state of their bodies. Riva starts their therapy with an interesting premise. Those patients believe in their perception. What they see must be true. By using VR, however, one can easily show that what we see can easily be not true. This 'de-synchronization' alters the patients perspectives. What they see can now be not true, and therefore look for different perspectives. This facilitates changing one's view of themselves, and helps them start treatment better.

Treatment of Fear of Public Speaking

Alexander De Foe et al [20] set out to treat Fear of public speaking using VR. Fear of public speaking (FOPS) is the debilitating anxiety and fear of speaking in public that it affects their social, academic and career lives. Patients with FOPS overestimate the negative evaluation of others, and underestimate their own proficiency. Public speaking situations become too jarring to face, and thus these individuals default to avoidance of the situations. In turn, such avoidance prevents them from growing and learning to deal with their fear, which leads to a lifestyle full of fear of facing the outside world. Traditionally, the most empirically supported FOPS treatments included cognitive-behavioral therapy, and controlled exposure to public speaking. Controlled exposure is a limited method because it relies on coordinating events, venues and suitable audiences, which delays therapy and increases costs. VR could potentially solve all these problems because it provides a repeatable and highly customizable experience to help patients in need whenever they need it.

For the study, the paper had three public speaking scenarios of varying difficulties, along with three unscripted speech topics in a single 60-minute session. The study hypothesized that VR exposure would elicit distress within each scenario. They hypothesized that repeated exposure would lead to learning of proficient management of the distress

during the speech itself. Finally, they hypothesised that repeated exposure would lead to proficient management of stress on the long run.

The experiment used the newer Oculus Rift headset, and a wrist-worn Fitbit Surge to record the heart rates of users. The experiment put 21 volunteers in three scenarios of varying difficulties: a small(10 agents) room, a medium (18 agents) room, and a large (46 agents) room. The agents were also randomized so that not all of them were attentive to the user. Some of them were programmed to look at the clocks on the walls or to be on their mobile phones. The users were requested to talk for 6 minutes about 3 different subjects, and were allowed 3 minutes rest between each subject. General distress, heart rates and opinions of users were collected and analyzed after.



Figure 2.8: The 3 scenarios posed by Alexander De Foe et al for their experiment on Treatment of fear of public speaking using Virtual Reality: (A) The small room, (B) The medium room, and (C) The large room.

As hypothesized, the scenarios elicited subjective distress, and their heart rates spiked, confirming their distress. Their second hypothesis regarding reduction of distress during the scenarios was not supported. This could be due to reduction of distress tolerance as the speech went on, or heightened awareness of symptoms, like sweaty palms, had users feel even more distressed. Their third hypothesis, though, about better management of distress after multiple sessions was supported, which is incredible because repeated exposure is simple to achieve using VR.

This study is limited by its small sample size. Also, the study only let the users rest for 3 minutes between each session, which is not enough time for emotional learning

and recuperation. Even though the users reported better management of distress between each scenario, it would have been ideal to have the users do speeches separated by multiple days to allow for resting.

2.2.5 Summary

In summary, there has been interest in VR to be used in purposes other than entertainment for decades now. However, progress on research for VR Therapy has been slow [14], because most research takes place at a time where HMDs were too expensive, and too cumbersome to be used comfortably by the patients. The studies have typically been small, and negative results were less likely to be reported. The gaps in the literature are large, where papers would study use cases of VR that were more known to work, like social anxiety, and leave out cases where the use of VR has unknown outcomes, like its use in the treatment of depression. VR has not been tested enough, and was used to investigate too few disorders, even though it has the potential to be a great assessment tool. The treatment trials have been small in size, mostly not pre-registered, and rarely carried out in accordance to the standards that are now expected in clinical research.

Chapter 3

System Design and Implementation

3.1 Design Goals



Figure 3.1: The target experience we aimed to reach [24].

Our goal with this project is to target a gap in research where VR has not been utilized as a communication tool where the psychiatrist can exist with the patient in VR. It has been used as a secondary tool in therapy, but not as the main way of achieve it. We wanted to create an immersive environment where psychiatrists and patients can feel like they actually share each other's space. Also, we wanted to augment the psychiatry sessions by providing experiences that would be too costly or inconvenient to achieve otherwise.

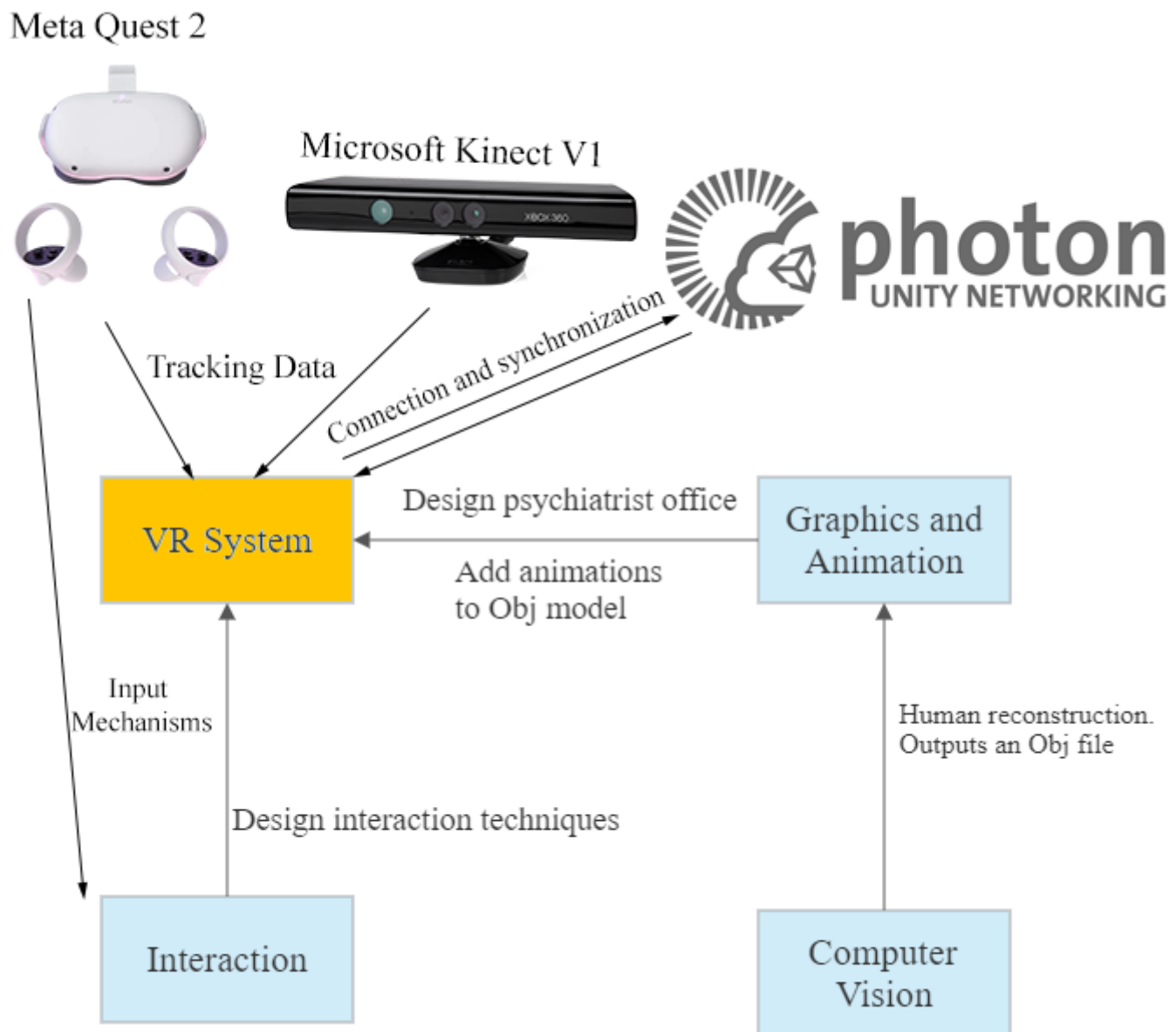


Figure 3.2: The system's block diagram.

3.2 System Overview and Design

3.2.1 General System Description

The system is intended to mimic a psychiatry session and bring it into the metaverse with all the great additions the virtual technologies bring.



Figure 3.3: Our system in action.

3.2.2 Application Flow

First, both users individually link their headsets to their PCs. Once the application is loaded, they will be asked if they have a Kinect or not. Then, the main room is loaded and

both users would be in the same room. They can see each other and talk to each other. The users can start recording their session, and engage together in various activities. They can share photos, videos, texts and audio, using various interaction techniques like grabbing objects and gesture controls. They can change the environment they are sitting in, from a normal room to a forest to a beach. Finally, an example "drill" was added for patients who suffer from anxiety or fear of public speech. These patients can try, along with their psychiatrists, to do a presentation in front of an audience on an actual stage.

3.2.3 System Modules

The system is split into 4 modules: Computer Vision, Graphics and Animation, Interaction and VR.

Computer Vision

This system is responsible for capturing the appearances of our users from a single RGB photo. From the photo, it creates a depth map to be used as avatars, bringing an authentic experience of actually existing in a virtual world as yourself.

3.2.4 Graphics and Animation

This system is responsible for bringing our users' to VR by having their actual faces generated and used in VR. Then, it uses lip-syncing with the input microphone audio to have the avatars' facial expressions change as they speak, giving more realism and immersiveness to the system.

Interaction

The interaction module uses principles of HCI to design and implement creative and efficient methods of interaction between the users using input data provided by the Quest 2 headset and controllers. It implements basic VR interaction like selection, manipulation, and locomotion. Other techniques are available to the users like gestures, remote interaction and grabbing using raycast from the controllers. Using these interaction techniques, the users can view and share different media content like video, audio, images, and text.

VR

The VR subsystem is responsible for bringing all the other modules together. It implements the means to exist and affect the world. The VR system is where we acquired the tracking of the Meta Quest 2 headset's tracking and combined it with the Kinect's

tracking to provide full animation of the avatars to match the users motions. It also implemented the means to give inputs to the UI in the world and interact with the objects by grabbing them and moving them around.

Moreover, this module includes the means to have both users connect through the network and be in the same room. Then, it allows their worlds to be synchronized. If a user affects the world, the same effect should happen at the other device. The users can talk to each other through the microphones, and communicate as if they were in the same room.

Finally, this module implements the auxiliary features that complement the rest of the project. Those features include different environments to have the session in, the ability to record the session, and a “therapy drill” of a presentation scene for the patient to try out.

These thesis focuses on the VR module, highlighting how it was designed and implemented.

3.3 Implementation Details

Since that our primary target is creating as much of an immersive experience as possible, the project is split into different features to enhance presence in VR.

3.3.1 Interacting With The World

For any of our systems to make sense, we need to have our user to be present in the scene, and for them to be able to interact with the world around them. We made use of the VR Interaction Framework [25] [asset](#) from Unity’s asset store, which helped us with some templates to work with.

Our main component is the XR Rig. XR is a term that stands for “extended reality” which combines both VR and AR systems. This component is the one to take the readings of the tracking of the controllers and headset, and translates them to coordinates that Unity can use. It also allows the user to interact with the world. Below highlighted are the most important components of the XR Rig:

The Input Bridge

Because the VR Interaction Framework is built to support multiples types of headsets and not just our Meta Quest 2 headset, the input bridge script detects the headset we are using, and provides a wrapper for detecting different inputs. Highlighted in figure 3.5 is the button layout of the controllers. The primary, secondary and thumbstick buttons provide boolean values: either a button is pressed or not. The inputs labeled “Axis1D”

are the trigger and hand grip inputs. These inputs provide a float value from 0 to 1. The inputs labeled "Axis2D" are the thumbsticks, which provide two float values: x and y values for the two dimensions the thumbsticks can move to. These float values allow us to not only detect an input, but also how far a button is pushed, which gives us the ability to make interactions that flow naturally. For example, we can use the float value for the trigger to detect how far the trigger is pressed to allow for a smoother animation of the hand. The harder the trigger is pressed, the more closed the avatar's hand would be.

The Player Controller

When wearing the headset, you can still move around your physical room, but we are not limited to just the bounds of our room. VR gives us the ability to be in extremely vast places from within our tiny rooms. Nevertheless, we need a way to move in said world. The player controller component on our XR Rig uses the previously mentioned Input Bridge to implement player locomotion. The script provides two forms of locomotion: continuous motion and teleportation. Continuous motion is to move around the scene by using the thumbsticks as you would move a character in a game. This type of locomotion, however, is usually considered for veteran VR users because it can easily induce heavy motion sickness. For beginner VR users, they get heavily immersed in VR that their body fully believes the experience. When using continuous locomotion, you move inside the application and the world moves past you, but your body is still static. This disconnect between what you visualize and believe versus what is actually happening is enough to make the body feel sick. It is similar to getting motion sickness in vehicles, only it is for the opposite reason. While you feel static in a vehicle even though you are actually moving, in VR you feel like you are moving even though you are static.

That is why we opted for using the simpler teleportation locomotion. Teleportation uses the thumbstick to trigger a reticle. You aim the reticle at the place that you want to teleport to and you would be instantly teleported there. Teleportation is considered a risk-free way of navigating in VR as it instantly translates a user to their desired destination. In doing so, it does not generate any optical flow that can be perceived as motion, thus reducing VR sickness. However, it has been argued that teleportation breaks presence since it lets users do something that does not exist in real life and its usage can also disrupt intended behavior of the system. Nevertheless, considering our target users, this compromise of a slight decrease in the feeling of presence is better than having users quit the application due to sickness.

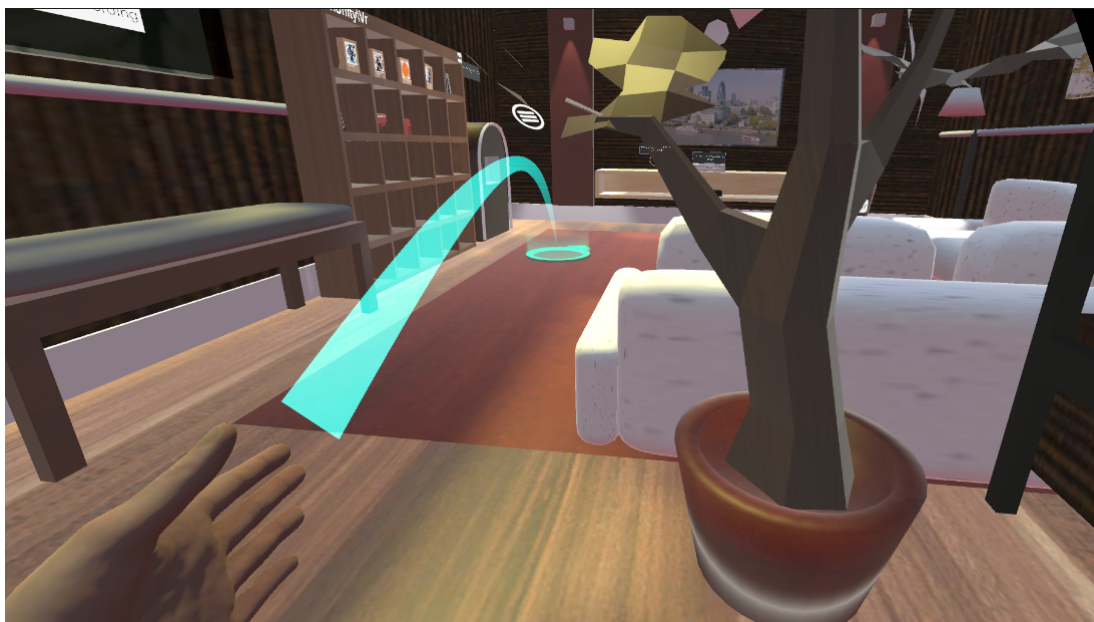
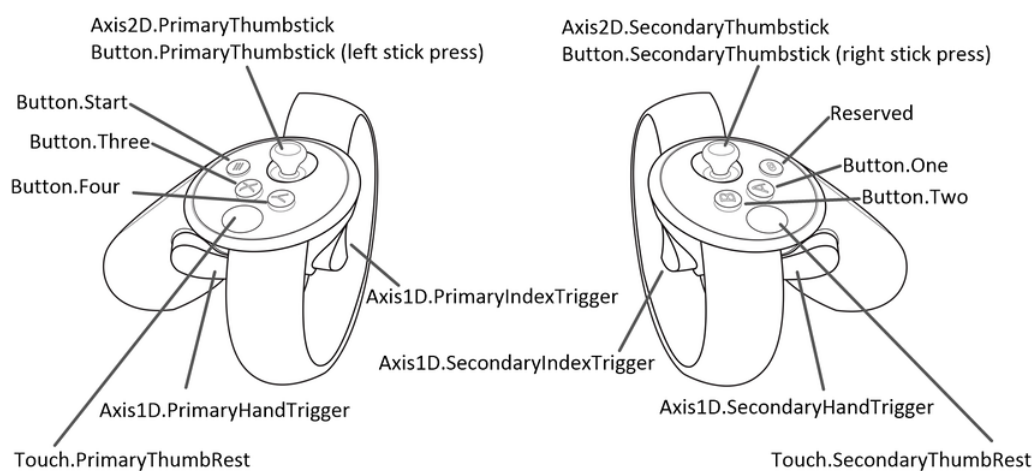


Figure 3.4: Teleportation reticle

Figure 3.5: The controllers' input layout. Image source from Unity's [Documentation](#).^[26]

The Camera and Tracking Space

We have three game objects as children to our rig: a camera and two controllers. Each game object has a tracked pose driver component. This component assigns a device to track to the game object, and updates its current transformations based on the tracked device. The XR rig has a script attached to it that calibrates the height of the user to match the avatar. On detecting a click on the the right thumbstick, the difference between the users current height and the desired height is calculated and added to the tracking data as an offset.

Attached to the right and left hand game objects are two more important components: the UI pointer component and the grabber component. The UI pointer script creates a small camera under one of the hands, and uses its raycast to detect where the hand is aimed and what UI element it is pointing at. When pressing the trigger on the controller, the script activates the UI element that the hand is pointing at. The grabber component detects collisions using Unity's physics system. If a grabbable object is currently colliding with the hands, and the user presses the grip, the grabbable objects position will follow the position of the hand giving the illusion of actually grabbing the object. It also has a remote grabbing component which functions similarly to how the UI pointer component functions. Using a raycast, the component detects if the user is aiming at grabbable component. If the component can be remote grabbed, it is teleported to the users hands, and begins to follow its transformations as usual.

3.3.2 Multiplayer

The "multiplayer" aspect of our project is one of its most important features. All implemented features had to be synchronized across both of our users. To implement real-time interaction between both users of our system, we made use of the Photon Pun 2 asset. Photon Pun is a unity package for multiplayer content that helped us with synchronizing the application states across both users. Even though Unity does provide its own multiplayer solutions, PUN provides an advantage that, along with a host/server system, it supports peer-to-peer communication while Unity's solutions do not. Peer-to-peer communication in our case meant for faster communication due to our application being on the smaller scale side of things. This extra speed along with extensive documentation and tutorials made PUN a better choice for us.

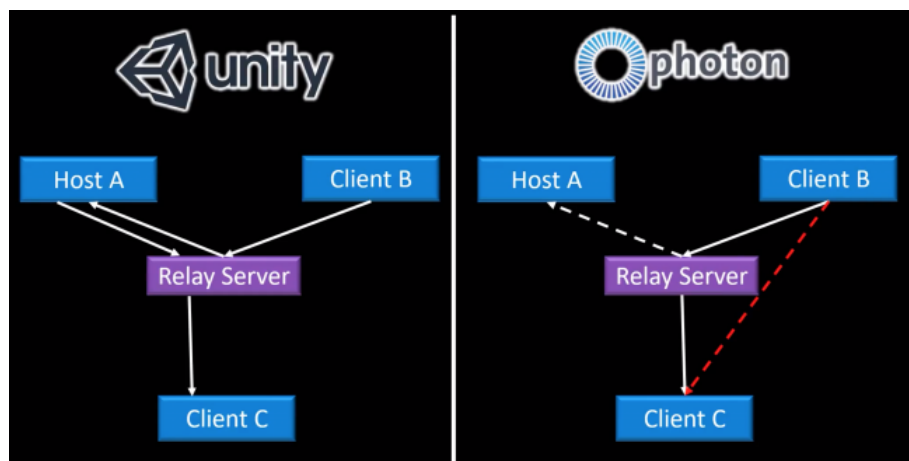


Figure 3.6: A comparison between Unity's networking system vs Photon. In Unity networking system, if client B wants to send a packet to client C, it has to go from the relay server to host A, then from host A back to the server and then to client C. For photon, client B can send a packet to client C through the relay server, but can also send it directly to client C through the peer-to-peer connection that Photon provides. [27]

PUN also allowed us to integrate the ability for both users to talk together using their microphones with the Photon Voice package. The audio is spatialized, meaning that audio output by users will be heard differently depending on where they are. If a person speaks to my right, I expect to hear it louder in my right ear. The 3D spatial audio helps us sell the idea of presence, because we hear people talking as naturally as we would do if we were just in the same room.

Objects that need to be synchronized had a photon view component added to them. Photon provides the ability to "raise events", which let us trigger functions on both clients using the application instead of just the local client that triggered the function. In the upcoming sections, we highlight how we implemented Photon to synchronize the features of the system across both users' clients.

3.3.3 Using Motion Tracking to Animate a Full-bodied Avatar



Figure 3.7: Motion Tracking in action

Motion Tracking is integral to the sense of presence in VR. Seeing your own body move the avatar you're controlling sells the idea that you are, in fact, present in the world you are interacting with. However, the Meta Quest 2 headset we were using is limited only to upper body tracking. So, an Xbox 360 (V1) Kinect was used to provide details about

the rest of the body. Even though it was more challenging to combine the tracking of both devices on a single 3D avatar, we opted for doing so as the tracking provided by the headset is much superior to the tracking provided by the Kinect. Using the Kinect was deemed sufficient as we require much finer control over our upper body than we do with our lower body. To synchronize the movement of the avatars for both users, a photon transform view component is added on the avatars, which syncs the transformations that both users apply to their avatars.

Tracking The Upper Body

The Meta Quest 2 headset provides excellent tracking of its controllers and the HMD itself using its inside-out sensors. Using the Unity asset **Final Ik**, the position of the controllers and the headset were mapped to the 3D avatar we were using. First, it maps to the hands and the head, and then the library uses inverse kinematics to calculate where the rest of the upper body should be. We assigned a VRIK component to the avatar, and mapped all the references to the avatar we are using. The different transform points correspond to their coordinates on the avatar. Notice in figure 3.8 for the VRIK component how for the left leg and right leg references, we added a dummy transform placed at the base of the avatar. Even though we do not need Final IK to move the legs, simply not assigning the variables would cause issues with the rest of the movement of the body. So, the transform just provides a sense of where the legs of the avatar are without altering the legs themselves. Then, you assign the head, right hand, and left hand targets in the solver part of the components. These targets are game objects which are children to each appropriate part game objects that track the absolute position of the headset and controllers. However, these game objects have a slightly different relative coordinates to the parent objects. Because otherwise, we could have cases where the camera is inside the head of the avatar. These extra game objects allow us to tweak the exact position we want our avatars to be in so that they look smooth and natural.

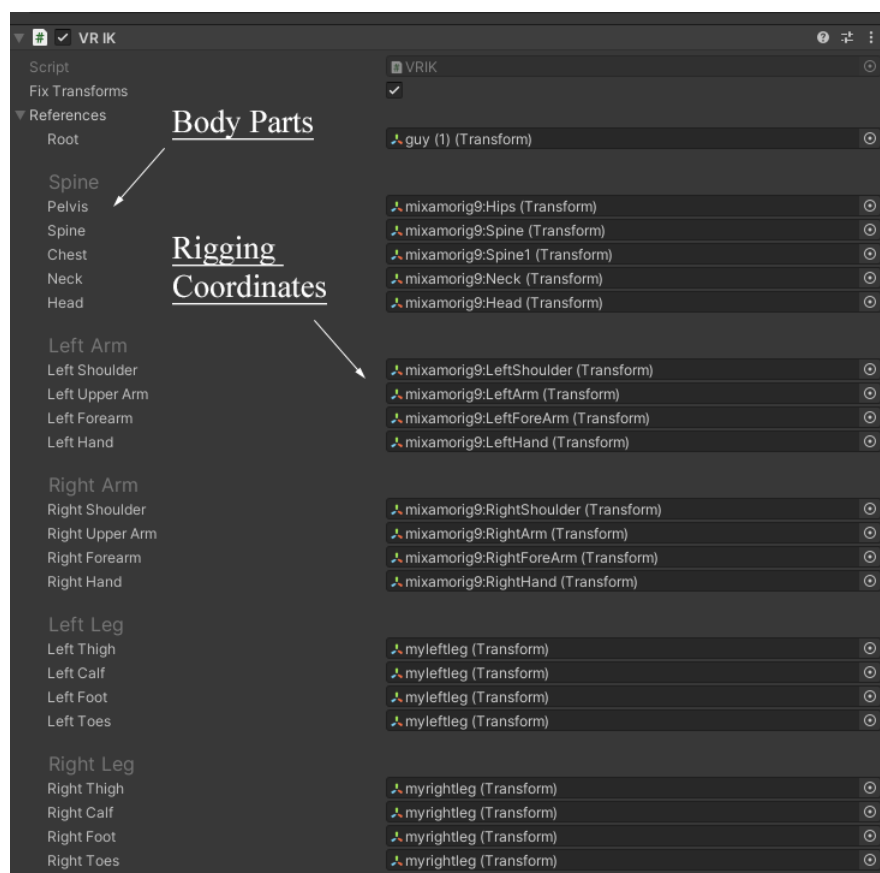


Figure 3.8: The VRIK component of the Final Ik asset

Tracking The Lower Body

To implement leg tracking with the Kinect, we made use of Shitao Fan's [28] [project](#). The Kinect identifies a player standing in front of it, and maps out a skeleton with all the joints matching the movements of the player. For our tracking to work, we need two components: the Kinect manager, and an avatar controller script on the avatar.

The Kinect manager script highlighted in figure ?? is responsible for calibrating and initializing the Kinect. It checks whether the Kinect is connected and powered up, and performs the necessary operations to acquire the tracking data provided by the Kinect. Some of the more important variables of script are:

- **Sensor Height** corresponds to how high the sensor is off the ground.
- **Sensor Angle** corresponds to the angle the sensor will be tilted to. Angle 0 is forwards without any tilt.
- **Min and Max User Distance** corresponds to how close and/or far the user needs to be for the Kinect to report its tracking data.

- **Smoothing** can be set to none, default, medium or aggressive. Smoothing can be helpful for the usual applications of the Kinect for games that can be viewed on an external screen. However, in our case where the Kinect is used to track the user to track their avatar, we need the movements of the avatar to be instantaneous. As a result, we set smoothing to be none.

The Kinect script matches those coordinates to the avatars we are using. By adding the avatar controller component to the same avatar we added the Final IK component to, we can move the legs using the readings from the Kinect.

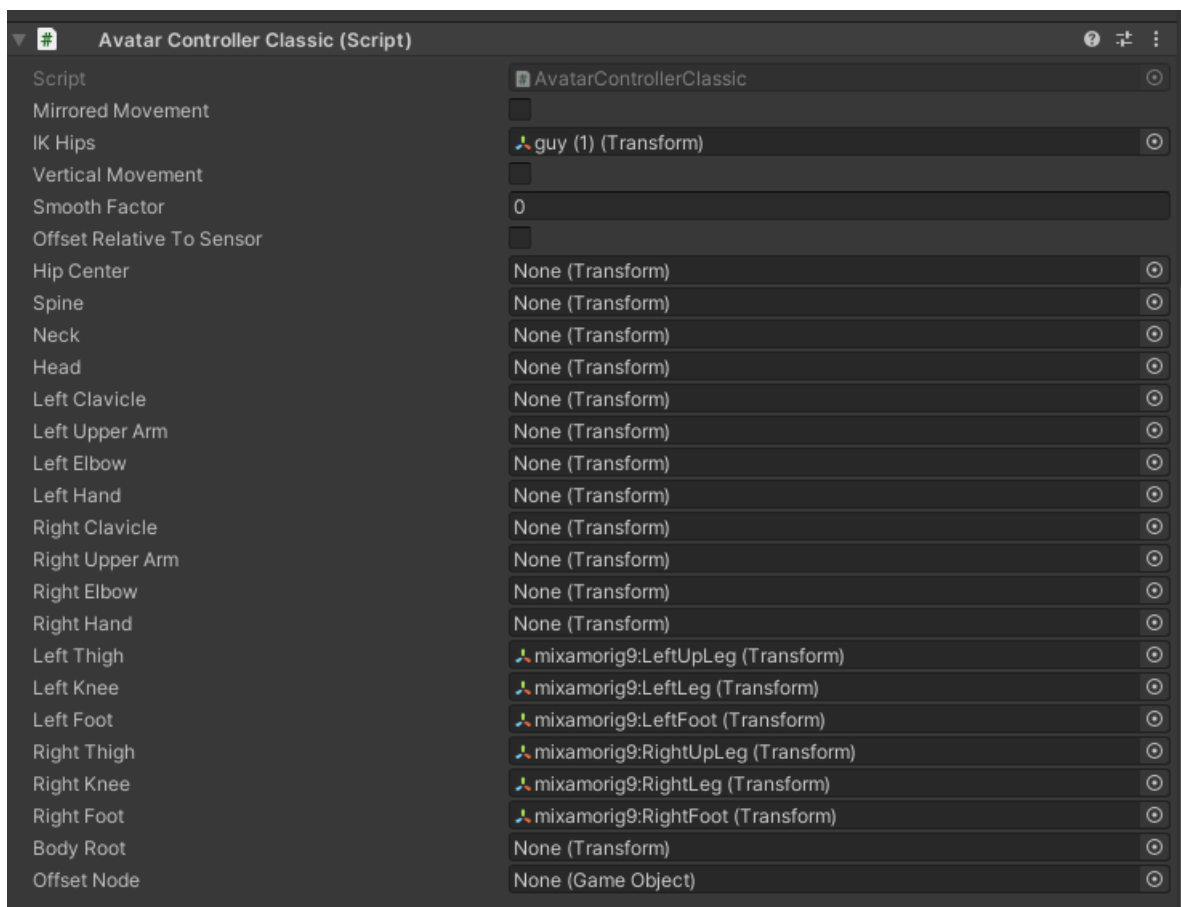


Figure 3.9: The Avatar Controller component.

Notice in figure 3.11 for the avatar controller component of the Kinect how we can leave the references empty here for the parts of the body we do not need. This script is different from Final IK's script in the sense that it does not need all the data to try and guess where the parts of the body *should* be. If a reference is missing, it does not alter how the present ones should move.

3.3.4 Sharing Videos

Videos are a fantastic method of conveying a lot of information in a short span of time. Psychiatrists can use it them to show their patients short movies that encapsulate a point to make. Even patients can use them to showcase their progress if they filmed themselves doing a task they could not achieve before.

To share videos, we made use of the Everreal Video Player package from the Unity asset store [29]. Unity does have its own video player, but this asset gave us some nicely designed UI and integration with the VR systems. The video player loads videos that are present locally on the device. We use photon to synchronize opening the same video files at the same time for both users. All controls are synchronized. Using photon, each UI element raises an event so that its function would occur on both devices.

The fact that files are loaded from local video files is both a positive and a negative. The positive aspect of it is that we forgo all problems with buffering and loading for weaker networks. The disadvantage is that the video file has to be present on both devices for it to load for both.



Figure 3.10: The Video Player.

Since we are in VR, we do not only have to watch 2D videos. In VR, we are in a 3D space and we can take advantage of that. The video player allows us to view 360 videos. They can use these videos for more immersive experiences. When a user chooses to change the rendering mode to 360 videos, we use photon to disable all the extraneous

elements in the room for both users so that they can be as immersed as possible in the experience.



Figure 3.11: 360 Videos.

3.3.5 Recording Sessions

Psychiatrists commonly record their sessions to listen to them afterwards for deeper analysis or just to remember what their patients had said in their sessions. It was also advised for the patients themselves to record the sessions for better recall and reflections on their sessions [30]. To record sessions, we record all of the output audio coming from the game, which includes the audio of the other person talking. We made use of Oleh Korotkov's [31] project as a starting point. Since that the output audio does not include the input audio from the user's microphone, we have to record that input separately and combine both recordings. To combine both audios, we made use of the NAudio library. The output does have a slight synchronization issue between the two audio files. However, this issue was outside of the scope of this project.

One hurdle we had to work through is that we could not have the microphone be used in two different scripts. For example, Photon was using the microphone for communication between the two users. In turn, the microphone would not work for recording. To overcome this issue, we create a microphone script that initializes the microphone only once. Then, assign this microphone to be used by other scripts. The difference this game object makes is that each individual script would not attempt to start the microphone

on its own and claim ownership of it. In doing so, we could use the same microphone input for multiple different scripts.



Figure 3.12: Recording UI

To start recording, the user picks their desired microphone. Then, they need to press the "Start Recording" button, which starts both scripts that record the game audio and the microphone audio. They need to specify the recording length because Unity's microphone recording needs to have knowledge of the recording length beforehand. When the "End Recording" button is pressed, we stop both recording scripts, which both output their separate recordings. Then, NAudio combines both recordings into a single file.

3.3.6 Different Environments

One of the benefits of being in a virtual world is that we are not confined to usual rooms. We implemented two other different environments: a forest and a beach. Each environment also has its own ambient noises to really sell the sense of presence in those environments. To reduce loading times in changing between the different environments, all the needed assets are initially loaded with the scene, and are spawned at very low Y-coordinates. When changing the environment, we simply raise the desired environment to the appropriate elevation, while lowering the current one to be practically out of view. This way, all the loading for the application is done only on startup. Also, we reset the position of both the users on changing the environment so that they spawn in a correct place and not fall of the environment. If the application bugs out, however, we always

check for the y-value of the user. If it falls below a certain threshold, we reset their position.

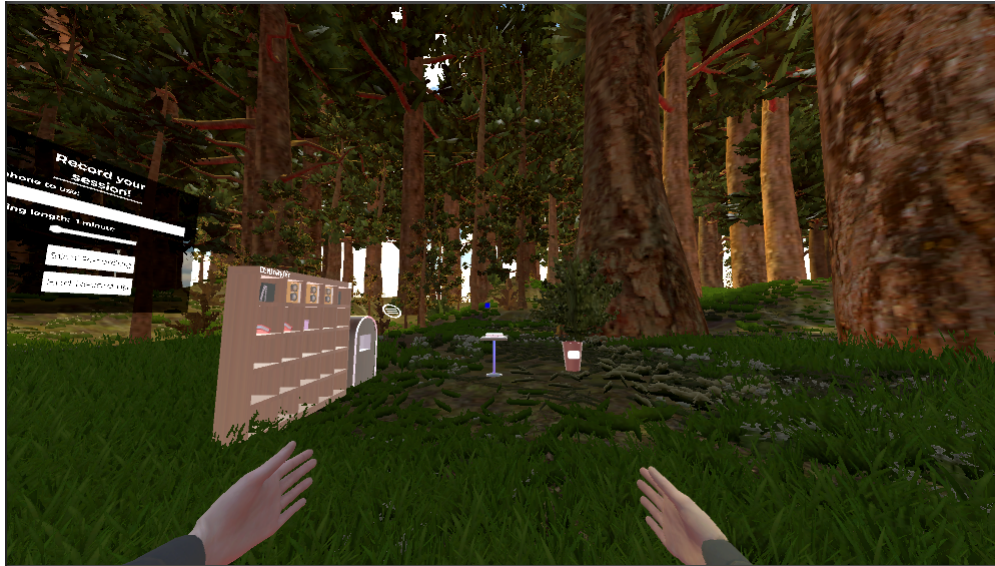


Figure 3.13: Forest Environment

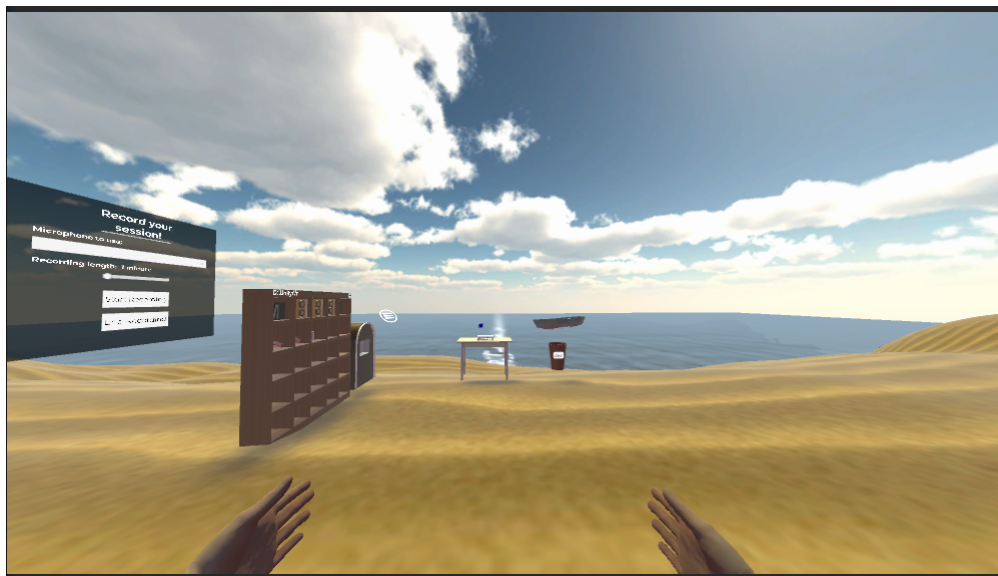


Figure 3.14: Beach Environment

3.3.7 Therapy Drills

Since we can simulate virtually any scenario in virtual reality, we proposed an example of a drill as a proof of concept. For anxious patients, or patients who suffer from fear of

public speaking or shyness, they can go through a presentation scenario in front of an audience. As established before, these drills are believable enough in VR, and the lessons learnt in these experiences carry over to the real world. We are addressing a gap here in research where the patient was usually alone in their experience in VR. As noted before, Alexander De Foe et al [20] carried out experiments that put patients with fear of public speaking in a virtual stage to do their speeches in VR. We have an advantage where the psychiatrist is also present with them in the scene. We propose that having an actual person as an avatar sells the idea more than the *other* avatars are also real. Also, having a person in the audience that the patients can feel safe giving their speech to might also help them with their anxiety.

We setup a stage scene with realistic avatars, all from mixamo.com, as an audience. The presenter is provided a whiteboard to use as they please, and their screen is duplicated for the psychiatrist to see. Finally, we simulate a mouse and keyboard for the presenter to interact with their computer without having to take off the headset.



Figure 3.15: Presentation Scene

Duplicating the screen

If the patient had prepared any material to display for their presentation, we needed a way for them to view it in VR. We achieve this by duplicating the patient's screen. Even though our application would be open up on the screen, minimizing the window will not affect its viewing in the headset. As a result, the user can open their favorite presentation application and view it in VR without ever taking off their headset.

We duplicate the user’s screen using the Desktop Duplication API provided by Microsoft for Windows, and Hecomi’s [32] project for reference. The API takes the currently rendered screen and puts it on a texture. This texture is then applied to a blank plane. The patient gets to see their screen inside the app with high fidelity at frames/second matching the screen’s refresh rate.

To share the screen for the psychiatrist, Photon allows sending basic types of data, but picture files are not supported. Photon is not inherently optimized to handle sharing data files. To capture a frame of the screen, we setup a camera directed at the monitor, and we use Unity’s ReadPixels call to capture the frame captured by that camera. To send the frame using photon, we serialize the frame into a byte array, then send the byte array to the other user. At the other client, we deserialize the array back into a jpg. This process is slow and we are only able to share a stream with a speed of a frame every 3 seconds. However, given the use case of just sharing presentation slides, our screen sharing system satisfies our need.

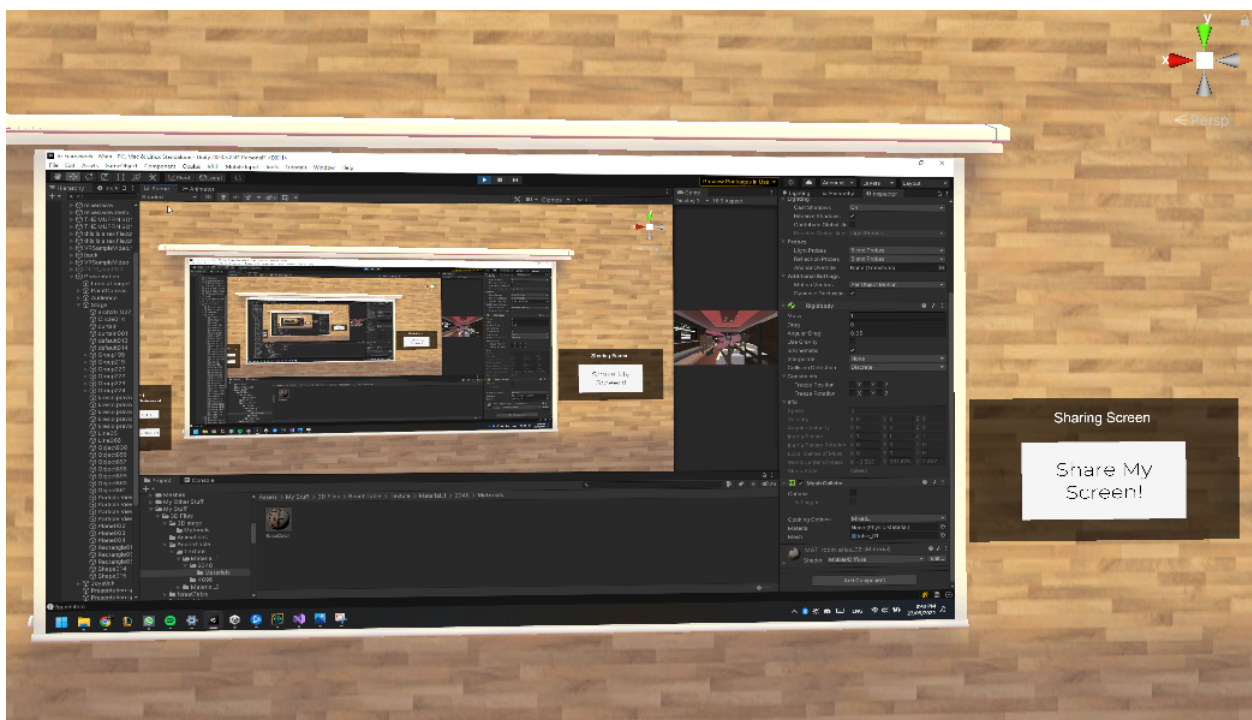


Figure 3.16: Sharing Screen

Simulating a mouse and a keyboard

Now that we have a way to view our screen in VR, we needed a way to control it. Taking the headset off to open up presentation slides would not only be cumbersome, but also breaks immersion.

To control the mouse, we setup a joystick. The horizontal and vertical angles of the joystick change the position of the mouse using the Cursor Control [33] asset from the

unity store. The movement of the cursor is slow to allow for accuracy, but holding the trigger on the controller bumps up the speed of the cursor to allow travelling of wider distances. While grabbing the joystick, you can click the primary button to simulate a left click, and the secondary button to right click.



Figure 3.17: The joystick to control the mouse

For the keyboard, we used the InputSimulator nuget package [34] to simulate keyboard strokes, which is a wrapper for the windows SendKeys api. For each button on the keyboard, a script component is added which assigns a keycode to be used by the package, and the keyboard buttons are activated with the UI ray.

3.4 Summary

In this chapter, we highlighted how we implemented our psychiatry in the metaverse system. Specifically, the VR subsystem of it. We explained how we tracked the user using their Meta Quest 2 headset and using the Microsoft Kinect. Once the user was in the virtual world, we explained how they would affect and be affected by the world around them. Then, we showed how these changes would be synchronized with the other user in real-time. Finally, we showed all the activities the users can engage in together. They can talk together, share videos, and change environments. The patient can also do a presentation to practice public speaking. During the presentation, the presenter can view their screen, and control their computer remotely using a joystick mouse and a keyboard. The presenter can also share their presentation to the other user, completing the presentation scene.

Chapter 4

Conclusion

4.1 Achievements

We have achieved an application that allows for two users to meet together and talk as if they were in the same room. Using VR, we made an immersive environment for users to be soaked in. We gave them the ability to even change these environments as they please. We gave the psychiatrists a lot of tools to work with during their sessions. They have a video player to share videos with their patient. They can record their session for later listening. We gave them the ability to be with their patients at all times if they wanted to, even during their patients' stressful situations.

And even though our main intention with our project was to have the psychiatrist and the patient always together, it does not limit our patient's ability to use the system alone. The patient can still use the system alone at anytime to sit in a quiet environment of their choosing. They can watch pre-made self-help videos or meditate in the environments. They can go through any drills we setup as many times as they need to until they feel comfortable taking on the real world.

4.2 Limitations

Even though our design goal was to have an image of two users *sitting* together, we could not have their avatars sit anywhere. That is because we do not know the actual rooms of both users, and we can not combine them in one room. If a user would sit on a chair in their room, they would most probably sit on air in VR.

Our user can still sit down, and calibrate the height of the avatar so that they are standing normally in VR, but we can imagine that the image of two users standing for an hour during therapy is not the expected experience.

Also, while the tracking provided by the Kinect V1 is passable, it is really limited. If a user were to rotate their body, the tracking would misinterpret the data. Moreover, the data provided by the Kinect is not quite in real-time, so there was a slight delay between a user raising their leg and it actually moving the avatar.

4.3 Future Work

Future work on this project is to mainly test out the project for its efficacy with actual psychiatrists and patients, and gather their opinions on the system.

Second, we discussed the limitations with the Kinect and how it generally poorly tracks the user. Future work on this project should consider either using a newer version of Kinect, or forgo it entirely and use a VR headset that provides full-body tracking natively. Finally, we propose to also add more drills and scenarios for psychiatrists and patients to go through to help different mental diseases. In doing so, it would flesh out the project to be a general purpose psychiatry alternative. Also, it should be considered to add difficulties to each drill so that the patients can go through these experiences gradually.

4.4 Summary

The world suffers from mental health issues, and not many ways to deal with those issues. And with the recent pandemic, the world has realized how underequipped it is to care for its mental health. We proposed and implemented a system for a psychiatrist to meet with their patients in VR. The system offers advantages over traditional therapy to save commuting time, and offer availability of psychiatrists to patients all over the world. In terms of online therapy over conferencing applications, it offers a sense of presence with the other person that simply can not be achieved through talking to a person through a screen.

Appendix

Appendix A

Lists

WHO	World Health Organization
VR	Virtual Reality
HMD	Head-Mounted Displays
HCI	Human-Computer Interaction
SoE	Sense of Embodiment
PTSD	Post-traumatic stress disorder

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